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(1) Research Program/Project/Study Title: Development of Seafood Paste Blue Crab (<i>Portunus pelagicus</i>) meat and Mud Clam (<i>Polymesoda erosa</i>)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	ROXAS CITY
		Province:	CAPIZ
		Region:	VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Foods	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Everyone knows that calcium is crucial for bone health. After calcium, the next most important substance for the health of your bones is phosphorous, which is crucial for the health of the bones and teeth. As a result, crab is an excellent additive to the diet of anyone who has a family history of osteoporosis and is at a higher risk of developing it. Eating crab for the high phosphorous content is essential, especially if you want to maintain an active lifestyle as you get older. The study utilized crab meat and mud clam, anato seed, anato oil, butter, garlic, salt, black pepper, and cayenne guided by the foregoing objective to determine the acceptability levels of four (4) treatments of seafood paste from blue crab meat and mud clam in terms of taste, color, aroma, mouth feel consistency and overall acceptability. The research used a completely randomized design (CRD). The development was done in 3 replicates. The utilization of crab meat and clam meat in making paste since crab meat contains high levels of omega-3 fatty acids, B12 and iron, vitamin C, and potassium in clam, thus making it a good substitute for shrimps, due to the above reasons the researchers intend to determine the level acceptability among treatments in terms of taste, color, aroma, texture, and appearance and determine if there is a significant difference in the acceptability among treatments. The quality of the product was determined by using sensory evaluation. The crab-clam paste had almost similar levels of acceptability in terms of taste, odor, color, texture aroma, and appearance regardless of the amount of crab meat and mud clam added to the formulated product.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Shrimps.

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Seafood products, including crustacean shellfish, are recommended for human diet due to their health-promoting characteristics (Skonberg and Perkins 2002). In terms of the amount of fat and the proportions of saturated, monounsaturated, and polyunsaturated fat, shellfish provide a healthful diet for humans (Dong 2001). Crab paste is a salt and liquor saturated product of fresh swimming crab, Portunus trituberculatus, well-known and sought-after in the coastal area of China (Ma, Xu, & Yang, 2008) A common sauce or paste that is used to flavor Asian food dishes. Crab paste is made from the meat of fresh crabs that is salt dried or salted and fermented to provide a soft, paste textured seasoning for foods. Similar to the Asian shrimp paste, crab paste is used to season soups, stews, curries, and rice dishes as well as serving as a marinade for various foods. Crab is packed with protein, which is important in building and maintaining muscle. Crab also contains a high level of omega 3, fatty acid, vitamin B12, and selenium, These nutrients play a vital role in improving general health while helping prevent a variety of chronic conditions. Improve heart health. The omega-3 fatty acids in crabs provide many benefits related to heart health. These important nutrients may help lower triglycerides, reduce blood clotting, and make it less likely that you'll develop an irregular heartbeat. Prevent anemia. Many of the nutrients found in crab, including vitamin B12 and folate, help reduce the risk of vitamin deficiency anemia. People with vitamin deficiency anemia do have enough healthy red blood cells and may experience fatigue or weakness as a result. Keep your brain strong. Research suggests that people who eat kinds of seafood, such as crabs, at least once a week have a reduced risk of dementia and omega-3 fatty acids found in seafood products.

Traditional crab paste production often uses a large amount of salt and wine to sterilize and inhibit bacteria to extend the shelf life, resulting in excessively high salt content and excessive wine flavor, which affects the flavor of crab paste and does not meet consumer demand for low-salt healthy food. At present, the per capita salt intake in my country has greatly exceeded the recommended intake of the World Health Organization, and pickled products are one of the important sources of salt intake for consumers. Studies have confirmed that a high-salt diet can cause diseases such as high blood pressure and coronary heart disease, and a moderate reduction in salt intake can significantly reduce the incidence of these diseases. The State Council has issued the “National Nutrition Plan (2017-2030)” action requirements to achieve the goal of reducing the national per capita daily salt intake by 20% by 2030. Therefore, the low salt curing process has attracted more and more attention from production and processing enterprises, but it also puts forward new higher requirements for the sterilization process.

A mud clam is Mollusca with two shells and has the shape of clams and shellfish but is larger in size. Mud clam lives and resides mainly on mangrove lands, riversides, coastal mudflats, and fish ponds of farmers in the provinces of the Philippines.

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Objectives:

This study will be conducted to development of Seafood Paste Blue Crab meat and Mud Clam.

Specifically, it aims to

1. Determine the acceptability levels of the four (4) treatments of seafood paste from blue crab (*Portunus pelagicus*) meat and mud clam in terms of appearance,taste, color, aroma, texture ,and overall acceptability.
2. Determine the most acceptable treatments in terms of appearance, taste, color, aroma , texture, and overall acceptability.
3. Determine the shelf life of the most acceptable treatment.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Materials

The following are the materials used in the preparation of seafood paste blue crab meat and mud clam frying pan, ladle, chopping board, measuring cups, measuring spoon, weighing scale.

Raw materials

The following are the raw materials used in the preparation of seafood paste blue crab meat and mud clam crab meat, mud clam meat, anato seeds, sugar, butter, cayenne, iodized salt, black pepper, annato oil.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Experimental treatment

Ingredients	Treatment A	Treatment B	Treatment C	Treatment D
Crab meat	100%	100%	100%	100%
Clam meat	none	25%	50%	75%

METHODS

Preparation of crabmeat

Wash the crab. Bring a large pan of water to a boil. Lower the fire, simmer for 12 mins per 1kg, then leave to cool. Remove the meat from the shell. Then set aside.

Preparation of Clams

The clams was depurated in chili pepper in water for 24 hours, then wash thoroughly in running water, and steamed for 30 min. the meat will be removed from the shell, remove the intestine organ and it will be chopped. Then set aside.

Cooking Procedure for each treatment

1. Heat the annatto oil and butter over low heat. Stir in the garlic and cook for about a minute.
2. Add the crab meat and clams and stir.
3. Season with salt, sugar, and pepper. Add cayenne to be spicy.
4. When everything is cooked through, remove from heat. Cool completely before storing them in bottles.

Formulation of mixture

The study used four (4) treatments with three replications: Treatment A (100%, crab meat,) Treatment B (100 %crab meat, mud clam 25%), Treatment C (100%, crab meat mud clam, 50 %) and Treatment D (100%, crab meat, mud clam 75%). Other ingredients were held constant.

Preparation of Seafood Crab paste meat and mud clam

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The ingredients of each formulated mixture detailed were weight accurately. Heat annato oil and butter over low heat fire. Saute in the garlic until golden-brown/ cook about a minute. add the crab meat then mud clam stir occasionally. Season with the salt, sugar, pepper and cayenne. When everything is cooked through, remove from heat. Cool completely before storing.

Shelf life of Paste

Shelf life being important criteria for seafood crab meat and mud clam, it is essential to estimate shelf life of the develop product. The developed seafood crab paste meat and mud clam gain highest score will subject to shelf life tests.

Sensory Evaluation of Paste

Sensory evaluation consists of judging the quality of food by a panel of 10 subject experts, and 100 consumers within Roxas City. The evaluation deals with analyzing and interpreting the qualities of the develop product .The scoring scale includes: 1 (dislike extremely), 2 (dislike very much), 3 (dislike moderately), 4 (dislike slightly), 5 (neither like or dislike), 6 (liked slightly), 7 (liked moderately), 8 (liked very much), 9 (liked extremely).

Experimental Design and Treatment

Experimental Design

The study used an experimental method of research. An experimental method is a method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to the mixture of the different samples (Calderron 2002)

Experimental Treatment

The study used four (4) treatments with three replications: Treatment A (100%, crab meat,) Treatment B (100%crab meat, mud clam 25%), Treatment C (7.5%, crab meat mud clam, 25%) and Treatment D (25%, crab meat, mud clam 75%). Other ingredients were held constant.

Data Collection

The data was gathered using a score containing the 9-point hedonic scale. It contains the rating of the panelist on color, odor, texture, taste, and general acceptability. The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Scoring of Variables

The criteria for evaluation as to sensory characteristics of the product was used 9-point hedonic scale in terms of color, odor, texture, and taste. The rating for general acceptability in a 9-point hedonic scale is 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor a dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin, 2006).

The score on each item was interpreted based on the result of the computation as to color, odor, texture, taste, and general acceptability using the following scoring interval and verbal interpretation.

Data Collection

The data was gathered using score containing the 9-point hedonic scale. It contains the rating of the panelist on color, aroma, texture, taste, and general acceptability. The data was gathered after every replication.

Scoring of Variables

The criteria for evaluation as to sensory characteristics of the product was 9-point hedonic scale in terms of color, odor, texture and taste. The rating for general acceptability in 9-point hedonic scale was: 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for

dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).
The score on each item was interpreted based on the result of the computation as to color, odor, texture, taste and general acceptability using the following scoring interval and verbal interpretation.

For Color

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Golden gate orange	8.12 – 9.0
8	Burnt orange	7.23 – 8.11
7	Alloy Orange	6.34 – 7.22
6	Red Orange	5.45 – 6.33
5	International orange	4.56 – 5.44
4	Orange soda	3.67 – 4.55
3	Red-orange	2.78 – 3.66
2	Yellow-orange	1.89 – 2.77
1	yellow	1.00 – 1.88

For Aroma

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
9	No fishy odor	8.12 – 9.0
8	With slight fishy odor	7.23 – 8.11
7	With a moderately fishy odor	6.34 – 7.22
6	With a Very much fishy odor	5.45 – 6.33
5	With an extremely fishy odor	4.56 – 5.44
4	With a Slightly Strong Fishy Odor	3.67 – 4.55
3	With a Moderately Strong Fishy Odor	2.78 – 3.66
2	With a Very Much Strong Fishy Odor	1.89 – 2.77
1	With an Extremely Strong Fishy Odor	1.00 – 1.88

For Texture

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Extremely firm	8.12 – 9.0
8	Very Firm	7.23 – 8.11
7	Moderately Firm	6.34 – 7.22
6	Slightly Firm	5.45 – 6.33
5	Neither Firm nor Soft	4.56 – 5.44
4	Slightly Soft	3.67 – 4.55
3	Moderately Soft	2.78 – 3.66
2	Very Soft	1.89 – 2.77
1	Extremely Soft	1.00 – 1.88

For Taste

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
9	No fishy odor	8.12 – 9.0
8	With slight fishy odor	7.23 – 8.11
7	With a moderately fishy odor	6.34 – 7.22
6	With a Very much fishy odor	5.45 – 6.33
5	With an extremely fishy odor	4.56 – 5.44
4	With a Slightly Strong Fishy Odor	3.67 – 4.55
3	With a Moderately Strong Fishy Odor	2.78 – 3.66
2	With a Very Much Strong Fishy Odor	1.89 – 2.77
1	With an Extremely Strong Fishy Odor	1.00 – 1.88

For General Acceptability

Scoring	Verbal interpretation/Response Categories	Score Interval
9	Like Extremely	8.12 – 9.0
8	Like Very Much	7.23 – 8.11
7	Like Moderately	6.34 – 7.22
6	Like Slightly	5.45 – 6.33
5	Neither Like nor Dislike	4.56 – 5.44
4	Dislike Slightly	3.67 – 4.55
3	Dislike Moderately	2.78 – 3.66
2	Dislike Very Much	1.89 – 2.77
1	Dislike Extremely	1.00 – 1.88

Statistical Tools and Analysis

The data was analysed using weighted mean and Analysis of Variance (ANOVA).
The weighted mean was used to determine the acceptability of the develop product in terms of color, aroma, taste, texture, and general acceptability of the product.
The analysis of variance was used to determine the significant difference among treatments in terms of color, aroma, taste, texture, and general acceptability.

For microbial analysis and shelf life:

The sample was sent to BFAR/SEAFDEC or any available accredited agency for analysis. The package was dependent on the requirement set by an accredited agency.

(12) Results and Discussions (This part contains a presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. The sequence shall be based on the objectives of the study. Tables are formatted without vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussion

A. In terms of taste

Table 1 shows the result of the acceptability of different Treatments in terms of taste.

As shown in the Table, Treatment A (100% crab meat) had a mean of 6.67, interpreted as moderate; Treatment B (100 % crab meat,25% clam meat) had a mean of 7.3, interpreted as like very much, Treatment C (100 % crab meat ,50% clam meat) had a mean of 7.50 interpreted as like very much and , Treatment D (100% crab meat and 75% clam meat) had a mean of 7.7 interpreted as live much.

Table 1. Acceptability of different treatments in terms of taste.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.67	6.67	6.53	6.67	6.67	Like moderately
B	7.2	7.3	7.63	7.2	7.3	Like very much
C	7.68	7.50	7.57	7.68	7.50	Like very much
D	7.47	7.7	7.53	7.47	7.7	Like very much

B. In terms of color

Table 2 shows the result of the acceptability of different Treatments in terms of color.

As shown in the Table, Treatment A (100% crab meat) had a mean of 6.67, interpreted as like moderately; Treatment B (100 % crab meat,25% clam meat) had a mean of 7.33, interpreted as like very much, Treatment C (100 % crab meat ,50% clam meat) had a mean of 7.55 interpreted as like very much and ,Treatment D (100% crab meat and 75% clam meat) had a mean of 7.64 interpreted as live much.

Table 2.Acceptability of different treatments in terms of color.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.83	6.70	6.47	20	6.67	Like moderately
B	7.57	7.2	7.23	22	7.33	Like very much
C	7.57	7.57	7.50	22.64	7.55	Like very much
D	7.50	7.67	7.76	22.93	7.64	Like very much

C. In terms of aroma

Table 3 shows the result of the acceptability of different Treatments in terms of aroma.

As shown in the Table, Treatment A (100% crab meat) had a mean of 6.76, interpreted as Like moderately; Treatment B (100 % crab meat,25% clam meat) had a mean of 7.42, interpreted as like very much, Treatment C (100 % crab meat,50% clam meat) had a mean of 7.40 interpreted as like very much and, Treatment D (100% crab meat and 75% clam meat) had a mean of 7.61 interpreted as live much.

Table 3. Acceptability of different treatments in terms of aroma.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.97	6.77	6.53	20.27	6.76	Like moderately
B	6.93	7.73	7.6	22.26	7.42	Like very much
C	7.5	7.27	7.43	22.2	7.40	Like very much
D	7.43	7.67	7.73	22.83	7.61	Like very much

D. In terms of texture

Table 4 shows the result of the acceptability of different Treatments in terms of texture.

As shown in the Table, Treatment A (100% crab meat) had a mean of 6.67, interpreted as moderate; Treatment B (100 % crab meat,25% clam meat) had a mean of 7.33, interpreted as like very much, Treatment C (100 % crab meat,50% clam meat) had a mean of 7.55 interpreted as like very much and, Treatment D (100% crab meat and 75% clam meat) had a mean of 7.58 interpreted as live much.

Table 4.Acceptability of different treatments in terms of texture.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.83	6.7	6.47	20	6.67	Like moderately
B	7.57	7.2	7.23	22	7.33	Like very much
C	7.57	7.57	7.50	22.64	7.55	Like very much
D	7.4	7.67	7.67	22.74	7.58	Like very much

E. In terms of appearance

Table 5 shows the result of the acceptability of different Treatments in terms of appearance. As shown in the Table, Treatment A (100% crab meat) had a mean of 6.50, interpreted as moderate; Treatment B (100 % crab meat,25% clam meat) had a mean of 7.33, interpreted as like very much, Treatment C (100 % crab meat,50% clam meat) had a mean of 7.41 interpreted as like very much and, Treatment D (100% crab meat and 75% clam meat) had a mean of 7.40 interpreted as live much.

Table 5.Acceptability of different treatments in terms of appearance.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.6	6.5	6.4	19.5	6.50	Like moderately
B	7.2	7.3	7.5	22	7.33	Like very much
C	7.37	7.47	7.4	22.2	7.41	Like very much
D	7.2	7.53	7.47	22.2	7.40	Like very much

ANOVA result of the significant difference in the acceptability of different treatments in terms of taste, color, aroma, texture and appearance

The results of ANOVA as reflected in Table 6 showed that there was no significant difference among treatments in terms taste, color, aroma, texture and appearance. This can be inferred that the evaluators accepted the formulated product at high level , and the level of acceptability increases as the amount of crab meat increases. Perhaps , the evaluators preferred the formulated product with higher amount of clam meat than with no crab meat added.

Table 6. Result of the Significant difference in the sensory characteristics among treatments in terms of taste, color, aroma, texture and appearance

Sensory Attribute	groupings	Sum of Squares	Df	Mean Square	F	Sig.
Taste	Between Groups	0.156	2	.078	.036	.965
	Within Groups	188.467	87	2.166		
	Total	188.622	89			
Color	Between Groups	.798	2	.399	.206	.814
	Within Groups	166.640	87	1.938		
	Total	167.438	89			

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Aroma	Between Groups	.422	2	.211	.111	.889
	Within Groups	156.067	87	1.794		
	Total	156.489	89			
Texture	Between Groups	1.667	2	.833	.398	.673
	Within Groups	181.933	87	2.091		
	Total	183.600	89			
Appearance	Between Groups	1.356	2	.678	.423	.656
	Within Groups	139.267	87	1.601		
	Total	140.622	89			
Acceptability	Between Groups	.622	2	.311	.155	.857
	Within Groups	175.033	87	2.012		
	Total	175.656	89			

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

Conclusion and Recommendations

The researchers utilized crab meat and clam meat in making paste since crab meat contains high levels of omega-3 fatty acids, and B12 and iron, vitamin C and potassium in clam , thus making it as a good substitute for shrimps

The level of acceptability of the formulated products varies as the amount of the clam meat varies but generally the developed was acceptable and similar when the different parameters was compared. Therefore ,the researchers encourage the utilization of crab and clam meat as substitute shrimp in making paste. It is recommended that this study be imparted to the community, especially those living near the coastal areas to maximize the potential of this raw material.

Microbial Analysis of Crab- clam meat paste is recommended and it will be followed.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Crab- clam meat paste

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Crab- clam meat paste

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should be include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention

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with 12-point space in between.)

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Kuley, E., Özoğul, >Fatih, Özogul, Y., & Olgunoglu, A. I. (2008). Comparison of fatty acid and proximate compositions of the body and claw of male and female blue crabs (*Callinectes sapidus*) from different regions of the Mediterranean coast. International Journal of Food Sciences and Nutrition, 59(7-8), 573–580. doi:10.1080/0963748070145120

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